Metropolis_Hastings_Algorithm

November 17, 2025

1 Metropolis-Hastings Algorithm Tutorial

This notebook provides a comprehensive introduction to the Metropolis-Hastings algorithm, a fundamental Markov Chain Monte Carlo (MCMC) method used for sampling from probability distributions.

1.1 1. Introduction to Metropolis-Hastings Algorithm

The Metropolis-Hastings algorithm is a Markov Chain Monte Carlo (MCMC) method for obtaining a sequence of random samples from a probability distribution from which direct sampling is difficult. The algorithm is particularly useful in Bayesian statistics and computational physics.

1.1.1 Key Concepts:

- **Markov Chain**: A sequence of random variables where the future state depends only on the current state
- **Proposal Distribution**: A distribution used to suggest new states
- **Acceptance Probability**: The probability of accepting a proposed state
- **Stationary Distribution**: The target distribution we want to sample from

[1]:
```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import norm, multivariate_normal
import time

# Set up plotting parameters
%matplotlib inline
sns.set_style('whitegrid')
plt.rcParams['figure.figsize'] = (12, 8)
plt.rcParams['font.size'] = 14
```

1.2 2. Basic Metropolis-Hastings Algorithm

The basic Metropolis-Hastings algorithm works as follows:

1. Initialize the chain with a starting point x
2. For each iteration t:
   - Generate a candidate x* from a proposal distribution q(x*|x)
   - Calculate the acceptance probability: = min(1, [p(x)*q(x /x)*] / [p(x)q(x*|x)])

- Accept the candidate with probability (set x = x*)
- Otherwise reject (set x = x)

3. Repeat for desired number of iterations

Where p(x) is the target distribution (up to a normalizing constant).

1.3 3. Implementation: Sampling from a 1D Distribution

Let's implement the Metropolis-Hastings algorithm to sample from a 1D distribution. We'll use a normal distribution as our target and a normal distribution as our proposal.

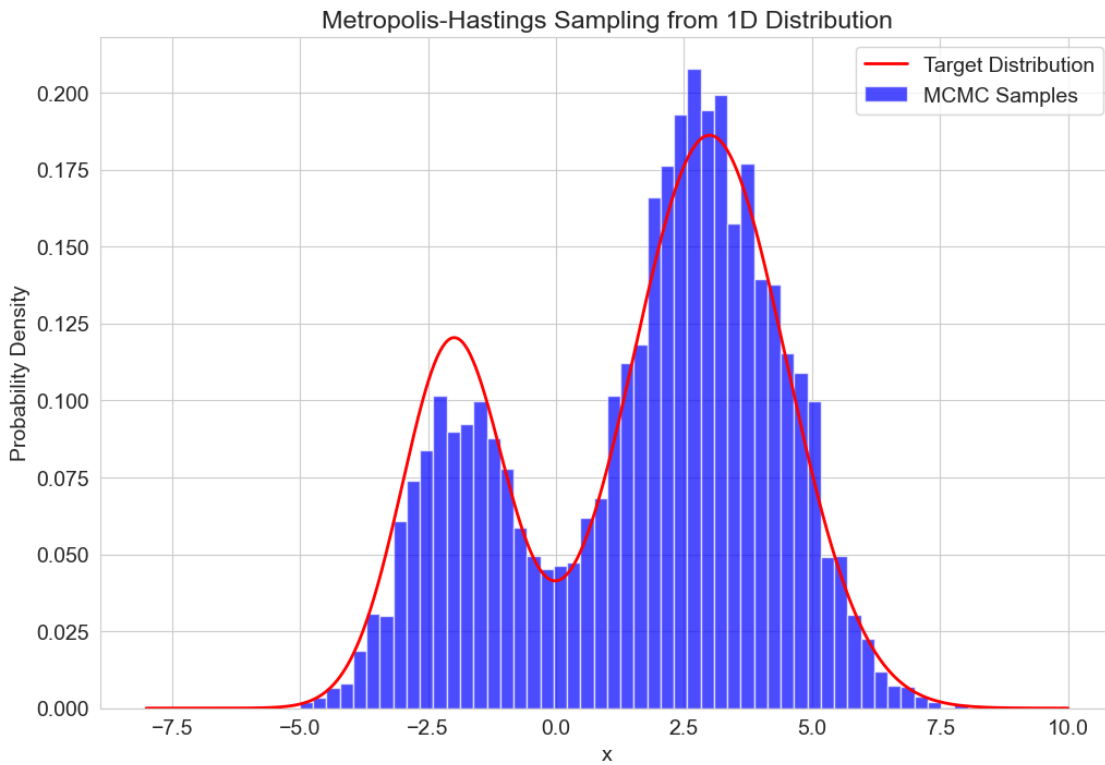
[2]:
```
def target_distribution_1d(x):
    """Target distribution: mixture of two normal distributions"""
    return 0.3 * norm.pdf(x, loc=-2, scale=1) + 0.7 * norm.pdf(x, loc=3,␣
 ↪scale=1.5)

def proposal_distribution(x, step_size=1.0):
    """Proposal distribution: normal distribution centered at current x"""
    return np.random.normal(x, step_size)

def metropolis_hastings_1d(target_func, proposal_func, initial_x, num_samples,␣
 ↪step_size=1.0):
    """Metropolis-Hastings algorithm for 1D distributions"""
    samples = np.zeros(num_samples)
    current_x = initial_x
    accepted = 0

    for i in range(num_samples):
        # Propose new state
        proposed_x = proposal_func(current_x, step_size)

        # Calculate acceptance probability
        # Since proposal is symmetric (normal), q(x*|x) = q(x|x*)
        # So acceptance probability simplifies to min(1, p(x*)/p(x))
        acceptance_prob = min(1, target_func(proposed_x) /␣
 ↪target_func(current_x))

        # Accept or reject
        if np.random.random() < acceptance_prob:
            current_x = proposed_x
            accepted += 1

        samples[i] = current_x

    acceptance_rate = accepted / num_samples
    return samples, acceptance_rate

# Run the algorithm
```

```
np.random.seed(42)
num_samples = 10000
initial_x = 0
step_size = 2.0

samples, acceptance_rate = metropolis_hastings_1d(
    target_distribution_1d, proposal_distribution, initial_x, num_samples,␣
 ↪step_size
)

print(f"Acceptance rate: {acceptance_rate:.2%}")
print(f"Sample mean: {np.mean(samples):.4f}")
print(f"Sample std: {np.std(samples):.4f}")
```

```
Acceptance rate: 68.90%
Sample mean: 1.7081
Sample std: 2.5431
```

1.4 4. Visualizing the Results

[3]:
```
# Plot the results
x_values = np.linspace(-8, 10, 1000)
y_values = target_distribution_1d(x_values)

plt.figure(figsize=(12, 8))

# Plot target distribution
plt.plot(x_values, y_values, 'r-', linewidth=2, label='Target Distribution')

# Plot histogram of samples
plt.hist(samples, bins=50, density=True, alpha=0.7, color='blue', label='MCMC␣
 ↪Samples')

plt.title('Metropolis-Hastings Sampling from 1D Distribution')
plt.xlabel('x')
plt.ylabel('Probability Density')
plt.legend()
plt.grid(True)
plt.show()
```

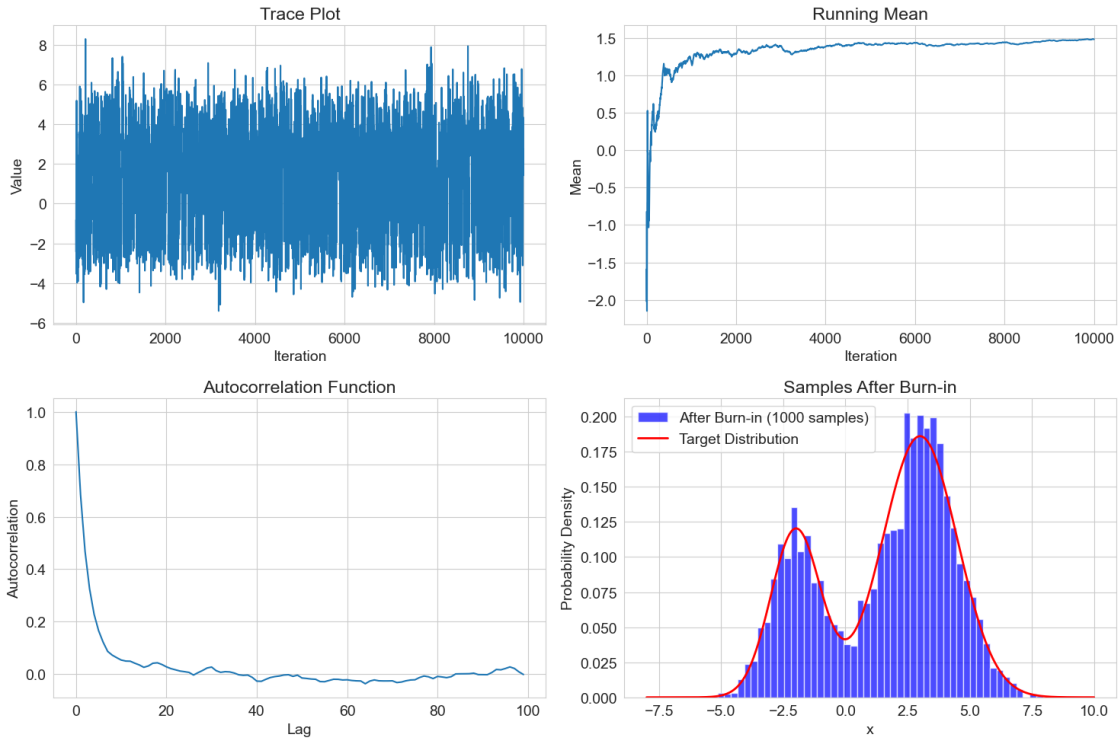


1.5 5. Analyzing Convergence and Mixing

Let's analyze the convergence of the Markov chain by looking at the trace plot and autocorrelation.

[10]:
```
# Create trace plot
plt.figure(figsize=(15, 10))

# Trace plot
plt.subplot(2, 2, 1)
plt.plot(samples)
plt.title('Trace Plot')
plt.xlabel('Iteration')
plt.ylabel('Value')
plt.grid(True)

# Running mean
plt.subplot(2, 2, 2)
running_mean = np.cumsum(samples) / np.arange(1, len(samples) + 1)
plt.plot(running_mean)
plt.title('Running Mean')
plt.xlabel('Iteration')
plt.ylabel('Mean')
plt.grid(True)
```

```python
# Autocorrelation plot
plt.subplot(2, 2, 3)
def autocorrelation(x, max_lag=100):
    n = len(x)
    autocorr = np.zeros(max_lag)
    x_centered = x - np.mean(x)

    for lag in range(max_lag):
        if lag == 0:
            autocorr[lag] = 1
        else:
            autocorr[lag] = np.sum(x_centered[:n-lag] * x_centered[lag:]) / np.
 ↪sum(x_centered**2)

    return autocorr

autocorr = autocorrelation(samples)
plt.plot(autocorr)
plt.title('Autocorrelation Function')
plt.xlabel('Lag')
plt.ylabel('Autocorrelation')
plt.grid(True)

# Histogram with different burn-in periods
plt.subplot(2, 2, 4)
burn_in = 1000
plt.hist(samples[burn_in:], bins=50, density=True, alpha=0.7, color='blue',␣
 ↪label=f'After Burn-in ({burn_in} samples)')
plt.plot(x_values, y_values, 'r-', linewidth=2, label='Target Distribution')
plt.title('Samples After Burn-in')
plt.xlabel('x')
plt.ylabel('Probability Density')
plt.legend()
plt.grid(True)

plt.tight_layout()
plt.show()
```

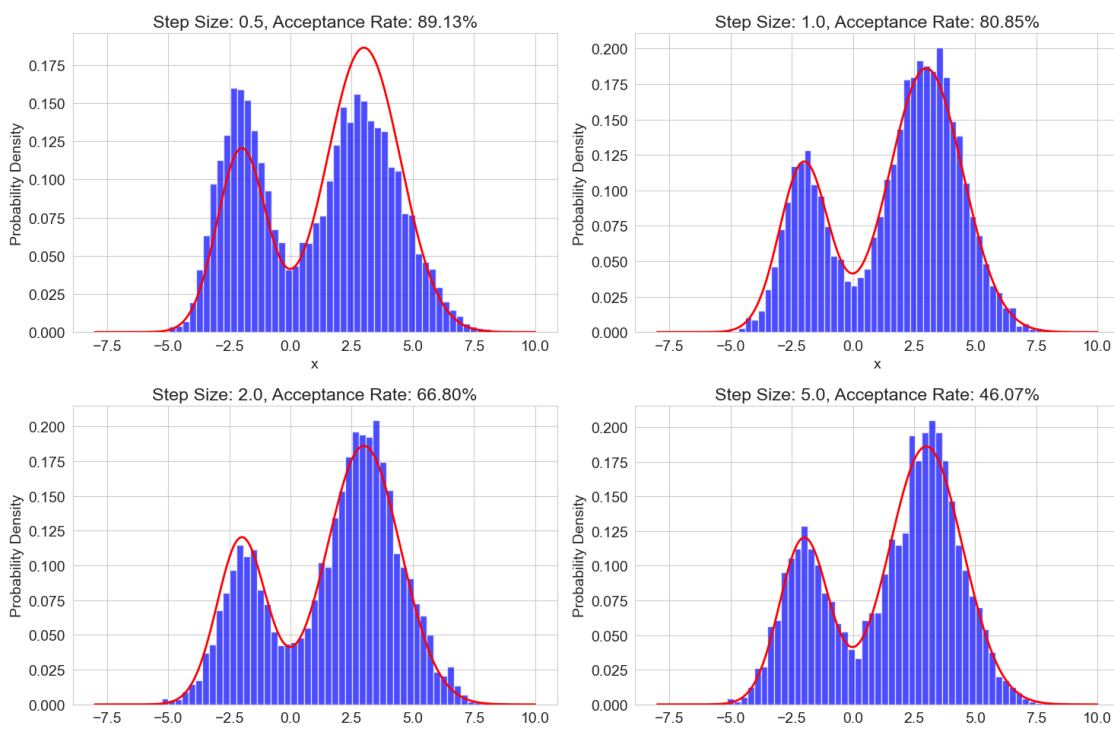


1.6 6. Effect of Step Size on Performance

The step size in the proposal distribution significantly affects the algorithm's performance. Let's compare different step sizes.

[5]:
```
# Compare different step sizes
step_sizes = [0.5, 1.0, 2.0, 5.0]
results = {}

for step_size in step_sizes:
    samples, acceptance_rate = metropolis_hastings_1d(
        target_distribution_1d, proposal_distribution, initial_x, num_samples,␣
↪step_size
    )
    results[step_size] = {
        'samples': samples,
        'acceptance_rate': acceptance_rate,
        'effective_sample_size': num_samples / (1 + 2 * np.
↪sum(autocorrelation(samples)[1:100]))
    }

# Plot results
plt.figure(figsize=(15, 10))
```

```python
for i, step_size in enumerate(step_sizes):
    plt.subplot(2, 2, i+1)
    plt.hist(results[step_size]['samples'], bins=50, density=True, alpha=0.7,␣
 ↪color='blue')
    plt.plot(x_values, y_values, 'r-', linewidth=2)
    plt.title(f'Step Size: {step_size}, Acceptance Rate:␣
 ↪{results[step_size]["acceptance_rate"]:.2%}')
    plt.xlabel('x')
    plt.ylabel('Probability Density')
    plt.grid(True)

plt.tight_layout()
plt.show()

# Print performance metrics
print("Performance Metrics for Different Step Sizes:")
print("-" * 60)
for step_size in step_sizes:
    print(f"Step Size: {step_size}")
    print(f"  Acceptance Rate: {results[step_size]['acceptance_rate']:.2%}")
    print(f"  Effective Sample Size:␣
 ↪{results[step_size]['effective_sample_size']:.0f}")
    print()
```



```
Performance Metrics for Different Step Sizes:
------------------------------------------------------------
Step Size: 0.5
  Acceptance Rate: 89.13%
  Effective Sample Size: 76

Step Size: 1.0
  Acceptance Rate: 80.85%
  Effective Sample Size: 200

Step Size: 2.0
  Acceptance Rate: 66.80%
  Effective Sample Size: 817

Step Size: 5.0
  Acceptance Rate: 46.07%
  Effective Sample Size: 1928
```

1.7 7. 2D Example: Sampling from a Bivariate Distribution

Now let's extend the algorithm to a 2D case, sampling from a bivariate normal distribution.

[6]:
```
def target_distribution_2d(x):
    """Target distribution: 2D normal distribution"""
    mean = np.array([1, 2])
    cov = np.array([[1, 0.8], [0.8, 1]])
    return multivariate_normal.pdf(x, mean=mean, cov=cov)

def proposal_distribution_2d(x, step_size=1.0):
    """Proposal distribution: 2D normal distribution centered at current x"""
    return np.random.multivariate_normal(x, np.eye(2) * step_size**2)

def metropolis_hastings_2d(target_func, proposal_func, initial_x, num_samples,␣
 ↪step_size=1.0):
    """Metropolis-Hastings algorithm for 2D distributions"""
    samples = np.zeros((num_samples, 2))
    current_x = initial_x
    accepted = 0

    for i in range(num_samples):
        # Propose new state
        proposed_x = proposal_func(current_x, step_size)

        # Calculate acceptance probability
```

```
        acceptance_prob = min(1, target_func(proposed_x) /␣
 ↪target_func(current_x))

        # Accept or reject
        if np.random.random() < acceptance_prob:
            current_x = proposed_x
            accepted += 1

        samples[i] = current_x

    acceptance_rate = accepted / num_samples
    return samples, acceptance_rate

# Run the 2D algorithm
np.random.seed(42)
num_samples_2d = 5000
initial_x_2d = np.array([0, 0])
step_size_2d = 1.0

samples_2d, acceptance_rate_2d = metropolis_hastings_2d(
    target_distribution_2d, proposal_distribution_2d, initial_x_2d,␣
 ↪num_samples_2d, step_size_2d
)

print(f"Acceptance rate: {acceptance_rate_2d:.2%}")
print(f"Sample mean: {np.mean(samples_2d, axis=0)}")
print(f"Sample covariance matrix:\n{np.cov(samples_2d.T)}")
```

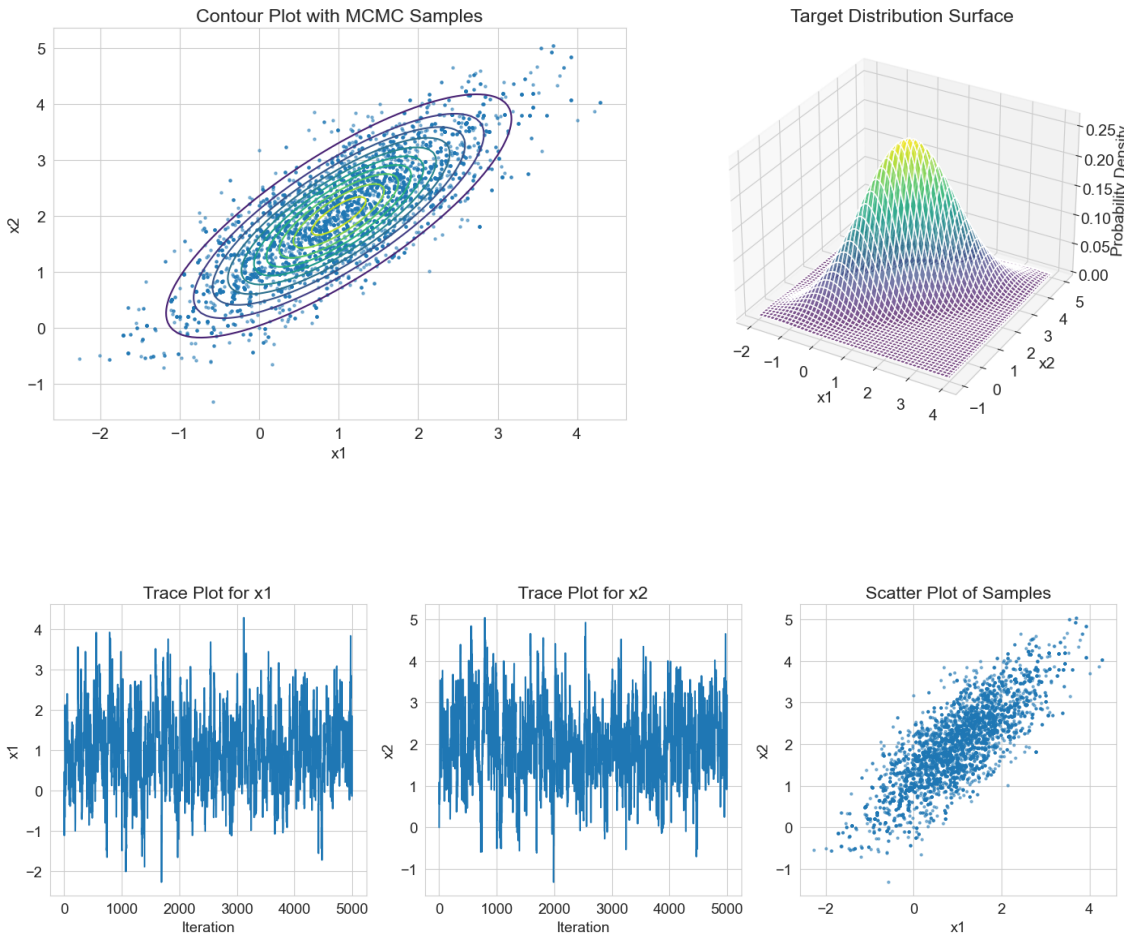
```
Acceptance rate: 39.56%
Sample mean: [0.98752191 2.00096823]
Sample covariance matrix:
[[0.93973118 0.74928676]
 [0.74928676 0.90817681]]
```

1.8 8. Visualizing the 2D Results

[7]:
```
# Create a grid for the target distribution
x = np.linspace(-2, 4, 100)
y = np.linspace(-1, 5, 100)
X, Y = np.meshgrid(x, y)
pos = np.dstack((X, Y))
Z = np.zeros_like(X)

for i in range(X.shape[0]):
    for j in range(X.shape[1]):
        Z[i, j] = target_distribution_2d(np.array([X[i, j], Y[i, j]]))
```

```python
# Plot the results
plt.figure(figsize=(15, 6))

# Contour plot with samples
plt.subplot(1, 2, 1)
plt.contour(X, Y, Z, levels=10, cmap='viridis')
plt.scatter(samples_2d[:, 0], samples_2d[:, 1], alpha=0.5, s=5)
plt.title('Contour Plot with MCMC Samples')
plt.xlabel('x1')
plt.ylabel('x2')
plt.grid(True)

# 3D surface plot
ax = plt.subplot(1, 2, 2, projection='3d')
ax.plot_surface(X, Y, Z, cmap='viridis', alpha=0.7)
ax.set_title('Target Distribution Surface')
ax.set_xlabel('x1')
ax.set_ylabel('x2')
ax.set_zlabel('Probability Density')

plt.tight_layout()
plt.show()

# Trace plots for each dimension
plt.figure(figsize=(15, 5))

plt.subplot(1, 3, 1)
plt.plot(samples_2d[:, 0])
plt.title('Trace Plot for x1')
plt.xlabel('Iteration')
plt.ylabel('x1')
plt.grid(True)

plt.subplot(1, 3, 2)
plt.plot(samples_2d[:, 1])
plt.title('Trace Plot for x2')
plt.xlabel('Iteration')
plt.ylabel('x2')
plt.grid(True)

plt.subplot(1, 3, 3)
plt.scatter(samples_2d[:, 0], samples_2d[:, 1], alpha=0.5, s=5)
plt.title('Scatter Plot of Samples')
plt.xlabel('x1')
plt.ylabel('x2')
plt.grid(True)
```

```
plt.tight_layout()
plt.show()
```



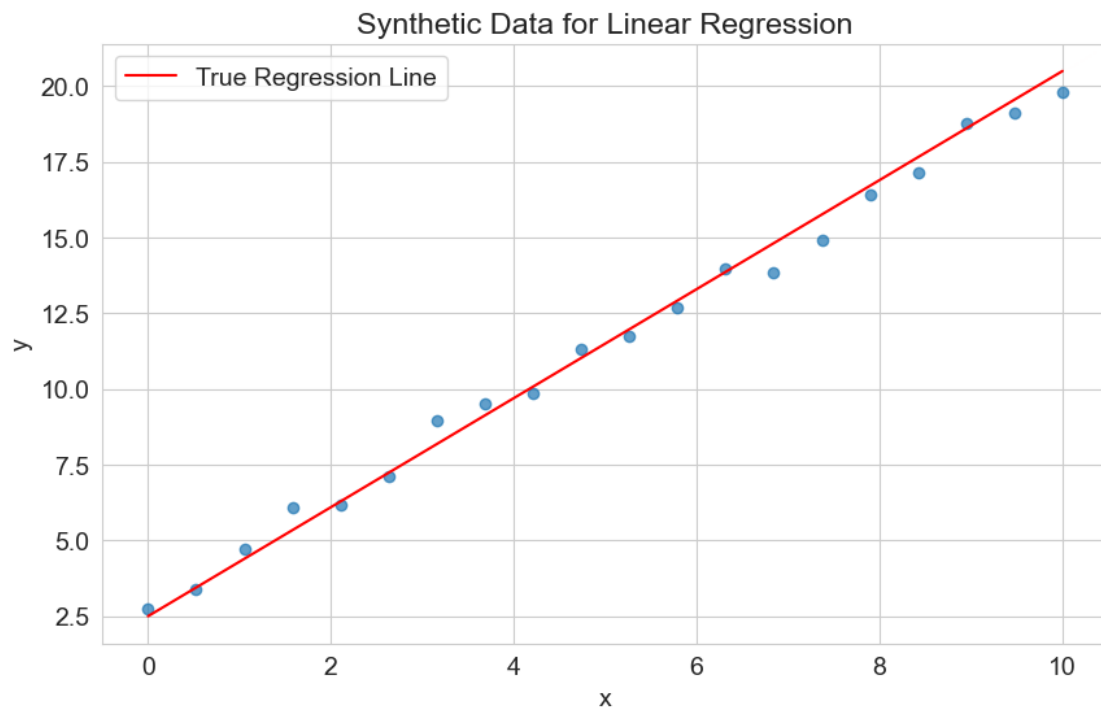
1.9 9. Advanced Example: Bayesian Linear Regression

Let's apply the Metropolis-Hastings algorithm to a Bayesian linear regression problem.

[8]:
```
# Generate synthetic data
np.random.seed(42)
true_intercept = 2.5
true_slope = 1.8
true_sigma = 0.5

x_data = np.linspace(0, 10, 20)
y_data = true_intercept + true_slope * x_data + np.random.normal(0, true_sigma,␣
↪size=len(x_data))

# Plot the data
```

```
plt.figure(figsize=(10, 6))
plt.scatter(x_data, y_data, alpha=0.7)
plt.plot(x_data, true_intercept + true_slope * x_data, 'r-', label='True␣
↪Regression Line')
plt.title('Synthetic Data for Linear Regression')
plt.xlabel('x')
plt.ylabel('y')
plt.legend()
plt.grid(True)
plt.show()

# Define the likelihood function
def likelihood(params):
    intercept, slope, sigma = params
    if sigma <= 0:
        return 0

    y_pred = intercept + slope * x_data
    residuals = y_data - y_pred

    # Normal likelihood
    log_likelihood = -0.5 * np.sum((residuals / sigma)**2) - len(x_data) * np.
↪log(sigma)
    return np.exp(log_likelihood)

# Define the prior distributions
def prior(params):
    intercept, slope, sigma = params

    # Normal priors for intercept and slope
    prior_intercept = norm.pdf(intercept, loc=0, scale=10)
    prior_slope = norm.pdf(slope, loc=0, scale=10)

    # Half-normal prior for sigma (positive only)
    if sigma > 0:
        prior_sigma = 2 * norm.pdf(sigma, loc=0, scale=1)
    else:
        prior_sigma = 0

    return prior_intercept * prior_slope * prior_sigma

# Define the posterior (up to normalizing constant)
def posterior(params):
    return likelihood(params) * prior(params)

# Metropolis-Hastings for Bayesian regression
def metropolis_hastings_regression(num_samples, initial_params, step_sizes):
```

```
    samples = np.zeros((num_samples, 3))
    current_params = initial_params
    accepted = 0

    for i in range(num_samples):
        # Propose new parameters
        proposed_params = np.array([
            np.random.normal(current_params[0], step_sizes[0]),  # intercept
            np.random.normal(current_params[1], step_sizes[1]),  # slope
            np.abs(np.random.normal(current_params[2], step_sizes[2]))  # sigma␣
↪(positive)
        ])

        # Calculate acceptance probability
        acceptance_prob = min(1, posterior(proposed_params) /␣
↪posterior(current_params))

        # Accept or reject
        if np.random.random() < acceptance_prob:
            current_params = proposed_params
            accepted += 1

        samples[i] = current_params

    acceptance_rate = accepted / num_samples
    return samples, acceptance_rate

# Run the MCMC
num_samples_reg = 10000
initial_params = [0, 0, 1]  # intercept, slope, sigma
step_sizes = [0.5, 0.5, 0.1]  # step sizes for each parameter

samples_reg, acceptance_rate_reg = metropolis_hastings_regression(
    num_samples_reg, initial_params, step_sizes
)

print(f"Acceptance rate: {acceptance_rate_reg:.2%}")
print(f"True parameters: intercept={true_intercept}, slope={true_slope},␣
↪sigma={true_sigma}")
print(f"Posterior means: intercept={np.mean(samples_reg[:, 0]):.2f}, slope={np.
↪mean(samples_reg[:, 1]):.2f}, sigma={np.mean(samples_reg[:, 2]):.2f}")
```




```
/var/folders/8x/k_m4pmb11437ktb_r6tjzt2c0000gn/T/ipykernel_71595/2620276245.py:6
9: RuntimeWarning: invalid value encountered in scalar divide
  acceptance_prob = min(1, posterior(proposed_params) /
posterior(current_params))
/var/folders/8x/k_m4pmb11437ktb_r6tjzt2c0000gn/T/ipykernel_71595/2620276245.py:6
9: RuntimeWarning: divide by zero encountered in scalar divide
  acceptance_prob = min(1, posterior(proposed_params) /
posterior(current_params))
```

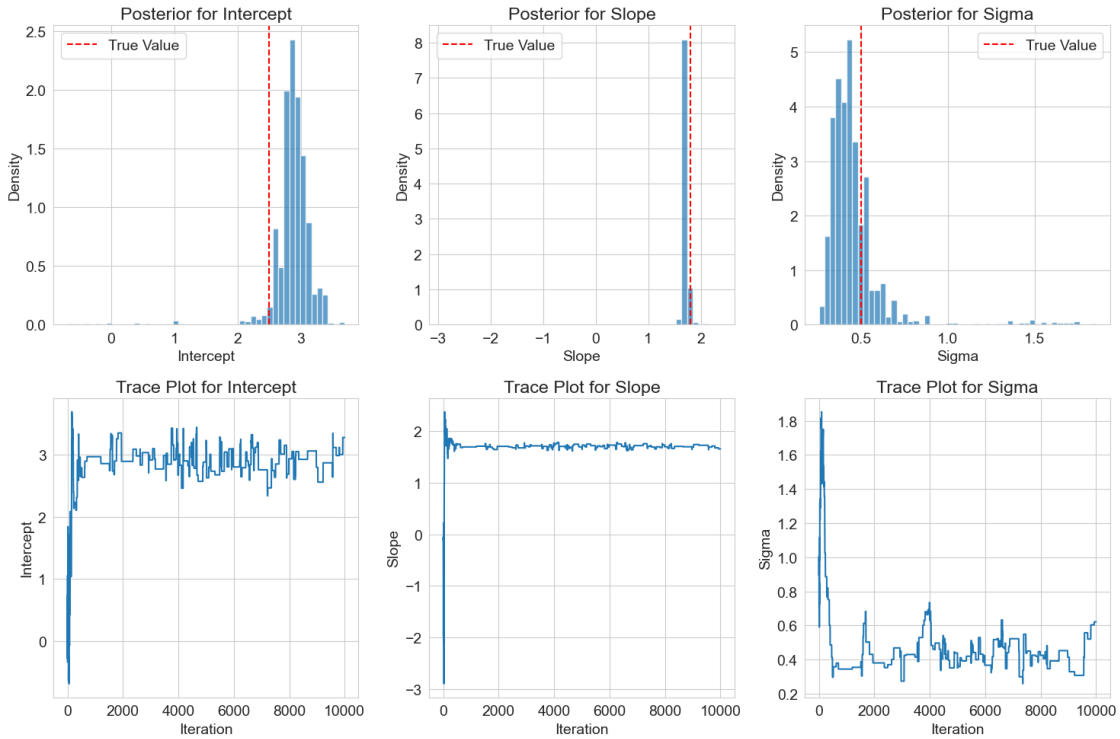
```
Acceptance rate: 2.15%
True parameters: intercept=2.5, slope=1.8, sigma=0.5
Posterior means: intercept=2.86, slope=1.70, sigma=0.46
```

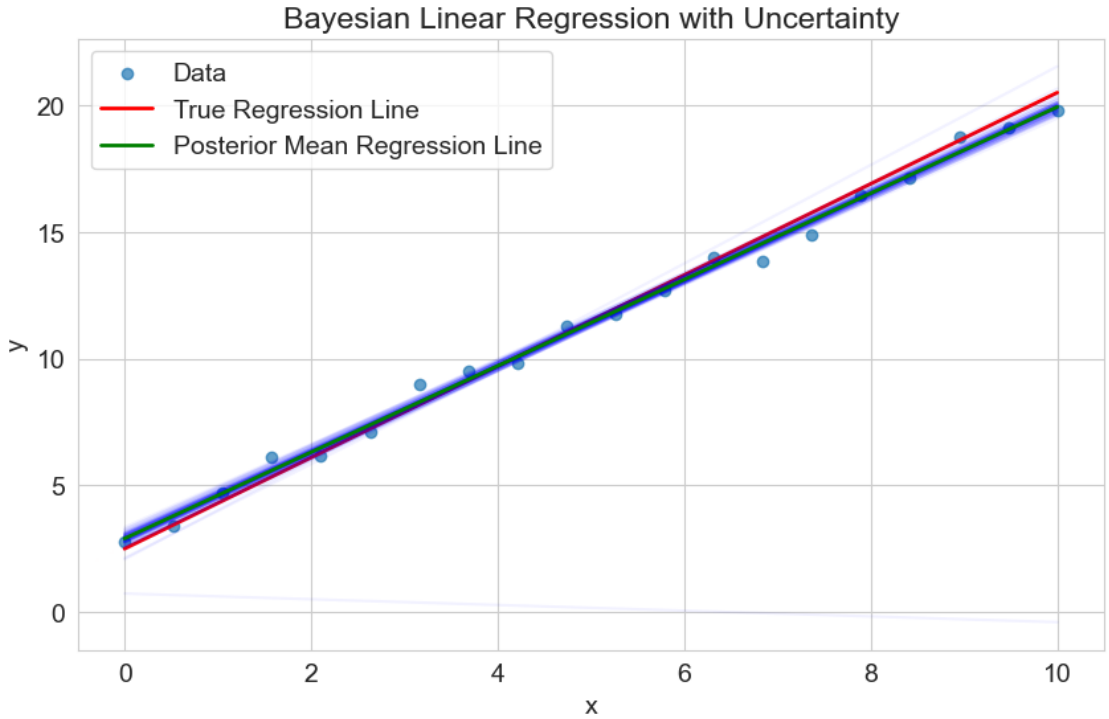
1.10 10. Visualizing the Bayesian Regression Results

[9]:
```
# Plot the posterior distributions
plt.figure(figsize=(15, 10))

# Posterior for intercept
plt.subplot(2, 3, 1)
plt.hist(samples_reg[:, 0], bins=50, density=True, alpha=0.7)
plt.axvline(true_intercept, color='r', linestyle='--', label='True Value')
plt.title('Posterior for Intercept')
plt.xlabel('Intercept')
plt.ylabel('Density')
```

```
plt.legend()
plt.grid(True)

# Posterior for slope
plt.subplot(2, 3, 2)
plt.hist(samples_reg[:, 1], bins=50, density=True, alpha=0.7)
plt.axvline(true_slope, color='r', linestyle='--', label='True Value')
plt.title('Posterior for Slope')
plt.xlabel('Slope')
plt.ylabel('Density')
plt.legend()
plt.grid(True)

# Posterior for sigma
plt.subplot(2, 3, 3)
plt.hist(samples_reg[:, 2], bins=50, density=True, alpha=0.7)
plt.axvline(true_sigma, color='r', linestyle='--', label='True Value')
plt.title('Posterior for Sigma')
plt.xlabel('Sigma')
plt.ylabel('Density')
plt.legend()
plt.grid(True)

# Trace plots
burn_in = 1000
plt.subplot(2, 3, 4)
plt.plot(samples_reg[:, 0])
plt.title('Trace Plot for Intercept')
plt.xlabel('Iteration')
plt.ylabel('Intercept')
plt.grid(True)

plt.subplot(2, 3, 5)
plt.plot(samples_reg[:, 1])
plt.title('Trace Plot for Slope')
plt.xlabel('Iteration')
plt.ylabel('Slope')
plt.grid(True)

plt.subplot(2, 3, 6)
plt.plot(samples_reg[:, 2])
plt.title('Trace Plot for Sigma')
plt.xlabel('Iteration')
plt.ylabel('Sigma')
plt.grid(True)

plt.tight_layout()
```

```
plt.show()

# Plot regression lines with uncertainty
plt.figure(figsize=(10, 6))
plt.scatter(x_data, y_data, alpha=0.7, label='Data')

# Plot true regression line
plt.plot(x_data, true_intercept + true_slope * x_data, 'r-', linewidth=2,␣
 ↪label='True Regression Line')

# Plot posterior regression lines
for i in range(0, num_samples_reg, 100):  # Sample every 100th iteration
    intercept = samples_reg[i, 0]
    slope = samples_reg[i, 1]
    plt.plot(x_data, intercept + slope * x_data, 'b-', alpha=0.05)

# Plot posterior mean regression line
mean_intercept = np.mean(samples_reg[burn_in:, 0])
mean_slope = np.mean(samples_reg[burn_in:, 1])
plt.plot(x_data, mean_intercept + mean_slope * x_data, 'g-', linewidth=2,␣
 ↪label='Posterior Mean Regression Line')

plt.title('Bayesian Linear Regression with Uncertainty')
plt.xlabel('x')
plt.ylabel('y')
plt.legend()
plt.grid(True)
plt.show()
```


Posterior for Intercept
True Value
Density
Intercept
Posterior for Slope
True Value
Density
Slope
Posterior for Sigma
True Value
Density
Sigma
Trace Plot for Intercept
Intercept
Iteration
Trace Plot for Slope
Slope
Iteration
Trace Plot for Sigma
Sigma
Iteration


Bayesian Linear Regression with Uncertainty
Data
True Regression Line
Posterior Mean Regression Line
y
x

1.11 11. Conclusion

In this notebook, we've explored the Metropolis-Hastings algorithm through several examples:

1. **Basic Implementation**: We implemented the algorithm for sampling from a 1D mixture distribution
2. **Performance Analysis**: We examined how step size affects acceptance rate and mixing
3. **2D Extension**: We extended the algorithm to a 2D bivariate normal distribution
4. **Practical Application**: We applied the algorithm to Bayesian linear regression

1.11.1 Key Takeaways:

- The Metropolis-Hastings algorithm allows sampling from complex distributions
- The choice of proposal distribution and step size significantly affects performance
- Convergence diagnostics like trace plots and autocorrelation are essential
- The algorithm is widely used in Bayesian statistics and computational physics

1.11.2 Practical Tips:

- Aim for an acceptance rate between 20-50% for optimal performance
- Use multiple chains with different starting points to check convergence
- Consider thinning (keeping only every k-th sample) to reduce autocorrelation
- Always perform convergence diagnostics before using the samples

The Metropolis-Hastings algorithm is a powerful tool for Bayesian inference and has been widely used in various fields including physics, statistics, and machine learning.